StudyShift: Machine Learning Lab Project Report

Team Members and Contributions

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  - Notebook 1: Mental Health Prediction
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  - Notebook 2: Multi-Model Academic Classification
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  - Notebook 3: Imbalanced Grade Prediction
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  - Notebook 4: Habit-Based Performance Regression

Project Repository

- **GitHub Link:** Link To the Project (https://github.com/Parth-Rathod-1729/IIITP_Sem5_ML_Lab_Project)

Project Details

Core Objective

- To predict student academic performance and mental health from multi-factor behavioral data, providing mathematically sound feature attributions to replace opaque risk labels.

Notebook 1: Mental Health Prediction (Random Forest)

- **What:** Predicts continuous `Mental_Health_Score` based on social media usage and behavioral factors.
- **How:** Compares a Ridge Regression baseline against a Random Forest Regressor pipeline.
- **Why:** Isolates non-linear dependencies, proving screen time and sleep duration are the fundamental drivers of student well-being (explaining 91.8% of variance).

Notebook 2: Academic Performance Classification (5 Models)

- **What:** Categorizes student performance into Low, Middle, and High tiers using the xAPI-Edu-Data dataset.
- **How:** Trains and evaluates Random Forest, SVM, Decision Tree, KNN, and Logistic Regression using standardized features.
- **Why:** Explores diverse mathematical decision boundaries (linear, distance, tree-based) to identify the optimal classifier for complex behavioral metrics.

Notebook 3: Student Performance Analysis (LR vs. RF)

- **What:** Predicts final letter grades (A through F) using demographic and academic history.
- **How:** Implements `ColumnTransformer` pipelines with stratified splitting and GridSearch, optimizing for the macro-F1 score.
- **Why:** Solves the fundamental problem of highly imbalanced target classes while preventing data leakage during cross-validation.

Notebook 4: Student Habits vs. Academic Performance

- **What:** Predicts continuous `exam_score` derived from daily habits, diet, and study patterns.
- **How:** Applies hybrid encoding (ordinal mapping + one-hot), standard scaling, and compares Linear, Ridge, Lasso, and Random Forest regressors.
- **Why:** Captures non-linear habit interactions and mitigates overfitting through L1/L2 regularization, establishing a robust baseline for continuous grade prediction.